Jihyung Kim

jihyung3@illinois.edu | +82 10-5753-5765 | Seoul, South Korea

Research areas: Quantum query complexity and Fourier analysis of Boolean functions, in particular the Fourier growth of quantum algorithms with few clean qubits (DQC_k) or few rounds of adaptivity.

Teaching areas: Deep Learning, Data Science, Data Structures and Algorithms, and AI literacy education.

Academic Employment

2026 – Part-time Lecturer, **Sahmyook University**, Department of Artificial Intelligence Convergence, Seoul, South Korea.

Education

2025 – 2026 **M.C.S.** in **Computer Science**, **University of Illinois Urbana-Champaign**.

2021 – 2023 **Ph.D. program** in **Computer Science**, **University of Central Florida** (withdrew).

2014 – 2021 **B.S.** in **Information Science**, **University of Arizona**.
(Minor in *Mathematics*)

Research Experience

2026 – Independent research in **Quantum Query Complexity**.
Fourier growth of DQC_k algorithms (in progress). Studying level-two Fourier bounds for quantum query algorithms with only k clean qubits and the query lower bounds they give, building on Girish, "Fourier Spectrum of Noisy Quantum Algorithms" (STOC 2026). Unpublished note (October 2026): a proposed proof of the level-two bound $d(2d-1)\cdot\sqrt{K}$ for d-query DQC_k algorithms, using the alternating-sign argument of that paper.
Adaptivity in quantum query algorithms (in progress). Studying the Fourier growth of algorithms with few adaptive rounds and the trade-off between rounds and parallel queries from Girish, Sinha, Tal, and Wu, "The Power of Adaptivity in Quantum Query Algorithms" (STOC 2024).

Graduate Research Assistant, **Socio-Technical Interaction Research (STIR) Lab**, University of Central Florida.

Contributed to a UX bootcamp project with teens (ages 13–17), co-designing effective online safety interventions while teaching UX design methods and industry-standard tools.

Undergraduate Research Assistant, **Future Reality Lab**, University of Arizona.
Researched an animated digital sketching interface for visualizing ideas and concepts in live presentation and conversation contexts.
Participated in the Sciences Designs Vertically Integrated Project (VIP) — advancing the UA Holodeck and developing health- and well-being-centered interactive experiences for everyday environments.

Industry Experience

Aug 2023 – Jan 2026 **Stealien Inc.** (cybersecurity firm), Senior Strategic Planner, Management Support Office, Seoul.
Led corporate strategy planning and business-environment improvement initiatives.
Identified and maintained external partnerships and collaborative relationships.
Planned new business initiatives and formulated new-market entry strategies.

Teaching

"AI Literacy and Problem Solving"
Course instructor at **Sahmyook University** (3 credits), Spring 2026.

"Data Structures and Algorithms"
Course instructor at **Sahmyook University** (3 credits), Spring 2026.

"Data Structures"
Course instructor at **Sahmyook University** (English-medium, 3 credits), Fall 2026.

"Introduction to Data Science"
Course instructor at **Sahmyook University** (English-medium, 3 credits), Fall 2026.

"Applied Deep Learning"
Course instructor at **Sahmyook University** (3 credits), Fall 2026.

"Foundations of HCI (CAP 3104)"
Graduate Teaching Assistant at **University of Central Florida**, Fall 2021 & Summer 2022.
50–60 undergraduates (Instructor: Mike McAlpin). Human-centered design principles, interaction modalities, design guidelines and heuristics, and interface evaluation techniques.

"Security in Computing (CIS 3360)"

Graduate Teaching Assistant at **University of Central Florida**, Summer 2022.
247 undergraduates (Instructor: Mike McAlpin). Applied security theory (legal and human factors), malware, intrusion patterns and tools, Windows/Unix/TCP-IP and application vulnerabilities, and detection, policy, and assurance.

"Program Analysis (COP 5021)"

Graduate Teaching Assistant at **University of Central Florida**, Spring 2022.
17 graduate students (Instructor: Ali Almalki). Static program analysis theory — data-flow analysis, abstract interpretation, and type-and-effect systems — with program analysis automation tools.

"Computer Understanding of Natural Language (CAP 6640)"

Graduate Teaching Assistant at **University of Central Florida**, Spring 2022.
51 graduate students (Instructor: Liqiang Wang). Advanced natural language processing course focused on developments of the preceding eight years.

"Introduction to Programming Language"

Volunteer Instructor at **Pima Community College** (Adult Basic Education for College and Career), Jan – Jun 2021.
Programming fundamentals — input/output, logic, decision structures, and loops — applying design thinking in open-ended projects with strategies such as Think-Pair-Share and pair programming.

"Introduction to Computer Science"

Classroom Volunteer / Teaching Assistant at **Ha:san Preparatory & Leadership School**, Aug – Dec 2017.
Taught Snap! programming to high school students; supported building a sustainable CS program through team-teaching that connected CS industry professionals with classroom teachers.

Technical Skills

Machine Learning: PyTorch, Keras, TensorFlow

Data Science: NumPy, pandas, scikit-learn, matplotlib, spaCy

Web / Mobile: Flask, React, Android Studio

Developer Tools: Git, UNIX command line

Military Service

Mar 2015 – Dec 2016 **Republic of Korea Army**, Sergeant (honorably discharged).